

29-5-2019 (E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Examinations
May-June 2019

Max. Marks:100

Class: Third year

Name of the Course: Operating System

Course Code: UCEC501

Duration: 3hrs

Semester: V

Branch: Comp

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Marks																								
Q 1 (a)	List and explain objectives and functions of operating system	10																								
Q 1 (b)	Distinguish between Layered V/S Monolithic structure of operating system	10																								
Q2 (a)	Explain process state transition diagram and types schedulers OR	10																								
Q2 (a)	Explain preemptive and non- preemptive scheduling algorithms	10																								
Q2 (b)	Consider the following set of processes, with the length of the CPU burst given in milliseconds. Processes: p1,p2,p3,p4,p5 Burst Time: 2,1,8,4,5 Priority: 2,1,4,2,3 The processes are assumed to have arrived in the order p1,p2,p3,p4,p5, all at time 0. 1) Draw three Gantt charts by using the following scheduling algorithms: FCFS, SJF and RR (quantum=2). 2) What is the turnaround and waiting time of each process 3) Which of the algorithms results in the minimum average waiting time (over all process).	10																								
Q3 (a)	Explain the Readers –Writers problem with structure process OR	10																								
Q3 (a)	Explain characteristics of modern operating system	10																								
Q3 (b)	Consider the following snapshot of a system. <table border="1"><thead><tr><th>Processes</th><th>Allocation (ABCD)</th><th>Max (ABCD)</th><th>Available (ABCD)</th></tr></thead><tbody><tr><td>P0</td><td>2001</td><td>4212</td><td>3321</td></tr><tr><td>P1</td><td>3121</td><td>5252</td><td></td></tr><tr><td>P2</td><td>2103</td><td>2316</td><td></td></tr><tr><td>P3</td><td>1312</td><td>1424</td><td></td></tr><tr><td>P4</td><td>1432</td><td>3665</td><td></td></tr></tbody></table> Answer the following questions using the banker's algorithm: a. Illustrate that the system is in safe state by demonstrating an order in	Processes	Allocation (ABCD)	Max (ABCD)	Available (ABCD)	P0	2001	4212	3321	P1	3121	5252		P2	2103	2316		P3	1312	1424		P4	1432	3665		10
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	which the process may complete. b. If request from process p1 arrives for (1, 1, 0, 0), can the request be granted immediately. c. If request from process p4 arrives for (0,0,2,0), can the request be granted immediately	
Q4 (a)	List files allocation methods and describe contiguous allocation method.	10
	OR	
Q4 (a)	Describe any two file accessing methods	10
Q4 (b)	Explain SSTF and FCFS disk scheduling algorithms	10
Q5 (a)	Explain File sharing and record Blocking	10
Q5 (b)	Compare Unix operating system and Mobile operating system (Android)	10
	OR	
Q5 (b)	Explain I/O buffering methods	10

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End Semester Examinations

May - June 2019

Max. Marks: 100

Class: T.Y. BTech

Name of the Course: Communication Systems

Course Code:

Duration: 3Hours

Semester: V

Branch: EXTC

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Marks
Q 1 (a)	A carrier wave $e_c = 4 \sin (2\pi \cdot 500 \cdot 10^3 t)$ is amplitude modulated by an audio wave $e_m = 0.2 \sin 3[2\pi \cdot 500 \cdot t] + 0.1 \sin 5[2\pi \cdot 500 \cdot t]$. (i) Determine the upper and the lower sideband frequencies, (ii) Modulation index of individual modulating signal, (iii) Amplitude of all sidebands, (iv) Total power in the sidebands, (v) Draw complete spectrum of AM signal. OR Explain in detail with neat sketch the third method for generation of AM SSB wave.	10 10
Q 1 (b)	Draw and explain operation of transmitter and receiver for adaptive delta modulation with neat sketch.	10
Q2 (a)	Explain the functioning of Foster-Seeley discriminator with neat diagram with the help of phasor diagram for FM demodulation. OR (i) Explain with neat circuit diagram operation of varactor diode direct PM modulator. (ii) Explain pre-emphasis and de-emphasis in FM with the help of circuit diagram.	10 10
Q2 (b)	Draw block diagram of FM receiver and explain its function in detail showing waveform at each stage.	10
Q3 (a)	State and prove sampling theorem for reconstruction of low pass signal. OR Explain with neat diagram operation of Adaptive Delta modulation (ADM) transmitter and Adaptive Delta modulation receiver. State advantages of ADM	10 10
Q3 (b)	Explain VSB transmission. Explain its application in TV system with neat diagram. State advantages of VSB.	10
Q4 (a)	Explain performance characteristics of radio receiver: (i) Sensitivity, (ii) Selectivity, (iii) Image frequency and its rejection in detail with neat diagram. OR Explain double spotting in radio receivers. For an AM broadcast superheterodyne receiver with RF and IF frequencies of 600 KHz and 455 KHz respectively, determine (i) Local	10 10

	oscillator frequency, (ii) Image frequency, (iii) Image frequency rejection ratio for a pre-selector Q of 100.	
Q4 (b)	Why AGC is required in radio receivers? Explain operation of delayed AGC with circuit diagram. State advantages of delayed AGC.	10
Q5 (a)	Explain with neat circuit diagram and waveform, generation and detection of PWM signal. OR Define noise figure. Derive the Friss formula for calculation of total noise figure for two amplifiers connected in cascade.	10 10
Q5 (b)	For the balanced modulator using FETs used for generation of DSBSC signal, show that the carrier is completely removed and the output voltage consists of both the sidebands.	10

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End Semester Examinations

May - June 2019

Max. Marks: 100

Class: TY B.Tech.

Name of the Course: Electromagnetic Engineering

Branch: ETRX

Course Code: UEXC501

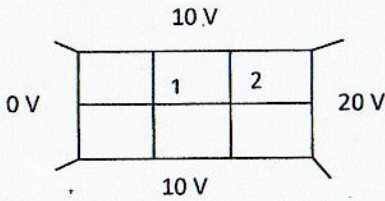
Duration: 3Hrs

Semester: V

Instructions:

- (1) All Questions are Compulsory
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Question No.		Marks
Q 1 (a)	Derive the boundary conditions for static electric and magnetic fields at a dielectric-dielectric boundary. OR Derive Maxwell's equations for time varying fields.	10 10
Q 1 (b)	In a homogeneous medium having $\mu_r=1$ and $\epsilon_r=15$, $\vec{E} = 50\pi e^{j(\omega t - \beta z)} \vec{a}_x$ V/m, $\vec{B} = \mu_0 H_m e^{j(\omega t - \beta z)} \vec{a}_y$ T. Find β and H_m if the wavelength is 2.38m.	05
Q 1 (c)	Using Maxwell's equations derive the Poisson's and Laplace's equation. OR State and explain Bio-Savart's law.	05
Q2 (a)	A plane wave propagating in free space has electric field intensity $\vec{E} = 200 \cos(\omega t - \beta z) \vec{a}_y$ V/m at 30MHz frequency. Find: i) Poynting vector ii) The time average power density in the wave. iii) The total power passing through the circular area of radius 5mm at $z=5$. OR Point charges of 50nC each are located at A(1,0,0), B(-1,0,0), C(0,1,0), and D(0,-1,0) in free space. Find the total force on a charge at A due to remaining three charges.	02 01 02 05
Q2 (b)	Derive the wave equation for a wave propagating in a conducting medium. OR Explain the concept of Skin Depth.	05 05
Q2 (c)	Derive the reflection coefficient for a perpendicularly polarized wave obliquely incident at a perfect dielectric. Hence, determine the reflection coefficient and the transmission coefficient at $\theta = 50^\circ$ at free space-dielectric boundary, dielectric constant being 3.5. OR What is polarization of EM waves? Explain different types of polarization.	08 02 10
Q3 (a)	Define degenerate modes, cut-off frequency, phase velocity and group velocity for an electromagnetic wave propagating in a waveguide.	05
Q3 (b)	Derive the expressions for the field components for TE_{mn} mode in rectangular waveguides. OR Derive the expressions for the field components for TE_{np} mode in circular waveguides.	10 10

Q3 (c)	A air filled rectangular waveguide is designed to propagate the dominant mode TE_{10} at 5GHz frequency. The cutoff frequency is 0.8 of the signal frequency. The ratio of the guide height to width is 2. Find the values of 'a' and 'b' for the waveguide. OR For an air filled circular waveguide with a radius of 5cm, operating at TE_{11} mode, i) Determine the cutoff frequency ii) Determine the guided wavelength for the operating frequency of 3GHz. iii) Determine the wave impedance in the guide. ($X'_{11}=1.841$)	05 01 02 02
Q4 (a)	Explain the principle modes of operation of Helical antenna and draw its radiation pattern. OR Derive the expression for the radiation resistance of a small loop antenna.	10 10
Q4 (b)	Explain the following terms related to the antenna: Radiation intensity, power gain, directive gain, and directivity. Find the average and maximum radiation intensity, U_{ave} and U_{max} respectively and the directivity, D if $U(\theta, \phi) = 4 \cos^2 \theta$, $\pi/3 < \theta < \pi/2$, $0 < \phi < \pi$.	06 04
Q5 (a)	Use the finite difference method to calculate the potentials at nodes 1 and 2 in the potential system shown in figure using iteration as well as band matrix method. 	10
Q5 (b)	Explain the method of moments and give its advantages and disadvantages.	10